Project Interim Report

On

Contact Directory Management
(Using Core Java & OOP Concepts)

RU-100-01-00012

Object Oriented Programming with Java

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Abstract

This project focuses on developing a Contact Directory Management System using Java and Object-Oriented Programming (OOP) concepts. The system is designed to store, manage, and manipulate contact details such as names and phone numbers efficiently. In today's digital world, managing contacts manually can be time-consuming and error-prone, so this system provides an organized and automated solution.

The project uses two main classes: Contact and Directory. The Contact class stores individual contact details, while the Directory class manages multiple contacts using an array. The system allows users to add new contacts, search for existing contacts using case-insensitive comparison, delete contacts, and display all stored records.

This project demonstrates real-world application of OOP principles like encapsulation, data management, and modular design. It also includes array manipulation techniques such as searching and element shifting during deletion.

Introduction

The Contact Directory Management System is a simple Java-based application that helps users maintain their contact details efficiently. It replaces traditional manual methods of storing contacts and provides a structured way to manage data.

Objectives

- Store contact records
- Search contacts easily
- Delete unwanted contacts
- Apply OOP concepts
- Reduce manual effort

Scope

This project is suitable for personal use or small-scale applications. It can be extended in future by:

- Adding database support
- Creating a GUI interface
- Storing additional details like email and address

System Requirements

Hardware:

- Computer with minimum 4GB RAM

Software:

- Java JDK
- IDE (VS Code / Eclipse)

Methodology

The system works in the following steps:

1. User adds contact → stored in array
2. User searches contact → system checks using `.equalsIgnoreCase()`
3. User deletes contact → array elements are shifted
4. User views all contacts

Flowchart

```
Start
  ↓
Display Menu
  ↓
Enter Choice
  ↓
┌──────────────┬──────────────┬──────────────┬──────────────┬──────────────┐
│ Add Contact  │ Search Contact│ Delete Contact│ Display All  │              │
└──────────────┴──────────────┴──────────────┴──────────────┴──────────────┘
  ↓
Perform Selected Operation
  ↓
Back to Menu
  ↓
End (if exit selected)
```

Class Diagram

Contact Class

- Variables: name, phone
- Methods: constructor, getters

Directory Class

- Variables: Contact[] contacts, count
- Methods:
  - addContact()
  - searchContact()
  - deleteContact()
  - displayContacts()

Implementation

Contact Class

Stores name and phone number using constructor and provides getter methods.

Directory Class

- Uses an array to store contacts
- **Add:** Inserts contact at next index
- **Search:** Compares names ignoring case
- **Delete:** Shifts array elements after deletion
- **Display:** Prints all contacts

Sample Input/Output

Contact added successfully

Searching for Rahul
Name: Rahul
Phone: 9876543210

Deleting Rahul...
Contact deleted successfully

Searching again...
Contact not found

Future Enhancements

- GUI interface using Swing/JavaFX
- Database integration
- Sorting contacts
- Add email/address fields

Gantt Chart (Write in Table)

Task	Time
Planning	Day 1
Design	Day 2
Coding	Day 3-4
Testing	Day 5

Conclusion

The project successfully demonstrates OOP concepts such as encapsulation and modular programming. It provides a functional system for managing contacts efficiently.

References

[1] H. Schildt, *Java: The Complete Reference*, McGraw-Hill
[2] Oracle, Java Documentation

💻 JAVA CODE (FINAL IMPLEMENTATION)

```
class Contact {
  private String name;
  private String phone;

  public Contact(String name, String phone) {
     this.name = name;
     this.phone = phone;
  }

  public String getName() {
     return name;
  }

  public String getPhone() {
     return phone;
  }
}

class Directory {
  private Contact[] contacts;
  private int count;

  public Directory(int size) {
     contacts = new Contact[size];
     count = 0;
  }

  public void addContact(String name, String phone) {
     if (count < contacts.length) {
        contacts[count++] = new Contact(name, phone);
        System.out.println("Contact added successfully");
     } else {
        System.out.println("Directory is full");
     }
  }

  public void searchContact(String name) {
     for (int i = 0; i < count; i++) {
        if (contacts[i].getName().equalsIgnoreCase(name)) {
           System.out.println("Name: " + contacts[i].getName());
           System.out.println("Phone: " + contacts[i].getPhone());
           return;
        }
     }
     System.out.println("Contact not found");
  }

  public void deleteContact(String name) {
     for (int i = 0; i < count; i++) {
        if (contacts[i].getName().equalsIgnoreCase(name)) {

           for (int j = i; j < count - 1; j++) {
```

```
                contacts[j] = contacts[j + 1];
            }

            count--;
            System.out.println("Contact deleted successfully");
            return;
        }
    }
    System.out.println("Contact not found");
  }

  public void displayContacts() {
    if (count == 0) {
        System.out.println("No contacts available");
        return;
    }

    for (int i = 0; i < count; i++) {
        System.out.println("Name: " + contacts[i].getName());
        System.out.println("Phone: " + contacts[i].getPhone());
    }
  }
}

public class Main {
  public static void main(String[] args) {

    Directory directory = new Directory(10);

    directory.addContact("Rahul", "9876543210");
    directory.addContact("Anita", "9123456789");

    System.out.println("\nSearching Rahul:");
    directory.searchContact("rahul");

    System.out.println("\nDeleting Rahul:");
    directory.deleteContact("Rahul");

    System.out.println("\nSearching again:");
    directory.searchContact("Rahul");

    System.out.println("\nAll Contacts:");
    directory.displayContacts();
  }
}
```